Rectified Flow for Chromosome Detection: Stable Coupling and Few-Step Inference

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***Abstract*—Chromosome karyotyping — the microscopic inspection of metaphase chromosomes for genetic diagnosis — remains a slow, labor-intensive, and observer-dependent cornerstone of clinical cytogenetics. Each metaphase cell contains roughly 46 densely packed chromosomes across 24 morphologically similar classes; automating this analysis requires a detector accurate and fast enough for routine clinical deployment. Conventional anchor-based detectors plateau on fine-grained intra-group discrimination, while diffusion-based detectors inherit the slow many-step inference and trajectory truncation errors of their image-generation ancestors, and small clinical datasets further destabilize training.**

We introduce *KaryoFlow*, a diffusion-based detector for chromosome karyotyping built on *Rectified Flow* (RF), which replaces the curved stochastic trajectories of DDPM with deterministic straight-line ODE paths. Three contributions target the specific difficulties of chromosome imaging. The RF paradigm itself yields the dominant accuracy gain, exceeding DiffusionDet by $+0.076$ mAP and matching Cascade R-CNN, with a controlled ablation attributing 94% of the gain to RF rather than to solver or step-count choices. We further characterize an *OT Diversity Collapse* that arises in the low-dimensional ($\mathbb{R}^4$) detection space and propose *Stochastic Coupling* via Sinkhorn transport, which restores coupling diversity and stabilizes training in the low-data regime. Finally, DPM-Solver++ with Top-K proposal pruning delivers four-step inference at clinical-grade latency, with a small but statistically significant precision advantage over the Heun baseline.

All claims are validated on two public chromosome datasets with multi-seed experiments, per-class AP analysis, and SOTA comparison, supporting diffusion-based detection as a practical paradigm for fine-grained medical imaging.

***Index Terms*—Rectified Flow, object detection, optimal transport, diffusion models, medical image analysis, chromosome karyotyping**

I. Introduction

a) Motivation.: Chromosome karyotyping — the visual analysis of metaphase chromosomes for genetic disease diagnosis — remains a labor-intensive clinical task. Each metaphase image contains about 46 tightly packed chromosomes across 24 classes (A1–Y), with severe class imbalance, fine-grained intra-group similarity, and frequent overlaps. Automating this process requires a detector that is both accurate and fast enough for clinical deployment, yet the three families of methods developed over the past decade each leave a measurable gap. Conventional anchor-based detectors (Cascade R-CNN, YOLOX-S) reach respectable throughput but plateau on the fine-grained 24-class discrimination that distinguishes the acrocentric C-group chromosomes from one another. Transformer detectors (DINO) lift accuracy but rely on multi-scale deformable attention and heavier backbones, unattractive when clinical data are scarce. Diffusion-based detectors (DiffusionDet) reformulate detection as iterative denoising from noisy boxes and offer a conceptually elegant fit for the structured-prediction nature of the task, but inherit the slow many-step inference and curved-trajectory truncation errors of their DDPM ancestors and are further destabilized by the small training sets typical of clinical imaging.

Rectified Flow [1] (RF) offers a principled remedy: by replacing the curved stochastic DDPM trajectory with a deterministic straight-line ODE path from noise to ground truth, RF enables few-step inference with low truncation error, especially valuable when per-image object density is high (~ 46 boxes) and errors compound across proposals. Applying RF to detection, however, raises three concrete questions: how should noise samples be coupled to ground-truth boxes when the prediction space is low-dimensional and the target count per image is large? Which ODE solver delivers the best accuracy–latency trade-off in the few-step regime? And how can training be stabilized on the imbalanced, small-scale corpora that characterize medical imaging? These three questions — coupling design, solver selection, and training stability — are the bottlenecks of RF-based dense detection in low-data regimes, and our three contributions address them systematically.

b) From scenario to method.: A typical metaphase spread contains roughly forty-six chromosomes across twenty-four classes, many touching or overlapping. The difficulty is uneven: C-group chromosomes (C6–C12) are morphologically similar, distinguished mainly by subtle banding patterns, while the Y chromosome is among the smallest, appears in only one copy in male samples, and has roughly $1{,}803$ training samples versus about 7,000 per autosome. A useful detector must therefore localize densely packed objects under few-step inference, train stably from an imbalanced small corpus, and run fast enough for interactive screening. These demands map onto our three ingredients: Rectified Flow supplies straight-line trajectories for few-step, low-truncation inference; Stochastic Coupling counteracts OT diversity collapse; and DPM-Solver++ with Top-K pruning converts trajectories into clinical-grade latency.

c) Contributions.: Building on the scenario-to-method mapping above, we make three contributions, each tied to a specific difficulty of chromosome detection and validated on two public datasets (Chromosome20240904, $1{,}540$ images; and 24 Chromosomes Object, $5{,}000$ images) with multi-seed experiments, per-class AP analysis, test-set evaluation, and an

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Code and models: https://github.com/[placeholder]

FPS benchmark.

The first difficulty is the accuracy ceiling of existing detectors on fine-grained 24-class chromosome imagery, where anchor-based designs struggle with dense packing and intra-group morphological similarity. Our *KaryoFlow* detector addresses this by adopting the RF training paradigm, achieving $+0.082$ mAP over the Euler baseline on 24 Chromosomes Object and $+0.017$ over DDPM on the original dataset. Because the A0$\rightarrow$A1 comparison changes several variables at once (DDPM$\rightarrow$RF, Euler$\rightarrow$Heun, $1 \rightarrow 4$ steps), a solver$\times$step disentanglement ablation attributes 94% of the gain to RF and only 6% to solver and step-count choices; AdaLN-Zero contributes null individually (Appendix A).

The second difficulty is a training pathology induced by optimal-transport (OT) coupling in the low-dimensional detection space. When $d = 4$ and $K \approx 46$, deterministic OT assignment collapses coupling diversity toward zero — a failure mode we characterize theoretically as *OT Diversity Collapse* (upper bound $\Delta H \leq \log K$, matched lower bound via Fano's inequality, empirically tight to within 0.03%) — and degrades training, especially when data are scarce. We propose *Stochastic Coupling*, which samples assignments from the Sinkhorn transport matrix rather than taking an argmax, restoring coupling diversity. The remedy is most valuable where clinical data are scarcest: on the smaller Dataset 1 it yields a large, highly significant mAP gain ($+0.034$, $p < 10^{-120}$) that diminishes with dataset size, and on both datasets it reduces within-run epoch-level mAP oscillation by $4.6\times$.

The third difficulty is inference latency: the curved, many-step trajectories of DDPM-based detectors are incompatible with interactive clinical screening. We deploy DPM-Solver++ [2] for four-step RF inference ($1.71\times$ speedup over Heun, mAP 0.859 ± 0.003 over three seeds), combined with Top-K pruning to reach 13.3–14.2 FPS. A per-image paired ablation reveals DPM-Solver++ yields a small but statistically significant precision advantage over Heun at matched step count ($+0.006$ per-image mAP, Wilcoxon $p < 0.001$, Table VII), refining FlowDet's [3] conclusion that higher-order solvers perform worse: at matched steps the higher-order solver is slightly *better*, not worse.

d) Novelty boundary.: Relative to FlowDet [3] (CFM with mini-batch OT, reports higher-order solvers perform worse), our novelty is: (i) the theoretical characterization of *why* mini-batch OT becomes a liability in low-dimensional structured prediction (Section III-C); and (ii) a solver$\times$step disentanglement showing that, at matched step count, the higher-order DPM-Solver++ is in fact slightly but significantly *better* than Heun (not strictly worse as FlowDet reports), while being $1.71\times$ faster in NFE. Relative to DeFloMat [4] (RF for medical detection, coupling as implementation detail), we provide the theoretical analysis of coupling design as a training pathology and the Stochastic Coupling remedy. Unlike OT-CFM [5] and multisample flow matching [6] (high-dimensional image generation, $d \sim 10^5$), our setting is low-dimensional ($d = 4$) with $K \approx 46$, where OT collapse is severe ($\Delta H/H \approx 0.69$). AdaLN-Zero [7] is reused as a standard implementation detail (Appendix A); DPM-Solver++ is adopted off-the-shelf, our contribution being the controlled disentanglement analysis with per-image paired statistical tests.

Figure 1 summarizes the framework.

II. Related Work

a) Diffusion-based object detection.: DiffusionDet [8] formulates detection as iterative denoising from noisy boxes using DDPM, but curved DDPM trajectories make few-step inference slow and truncation-error-prone. DiffuBox [9] extends the diffusion paradigm to 3D object detection via point-diffusion refinement of coarse proposals, but the 3D point-diffusion setting differs substantially from our 2D noise-box-to-GT formulation. FlowDet [3] applies Conditional Flow Matching with mini-batch OT coupling and reports higher-order solvers perform worse, but does not analyze *why* mini-batch OT becomes a liability in low-dimensional structured prediction. DeFloMat [4] uses Rectified Flow for medical detection but treats coupling as an implementation detail. Our work closes these gaps with a theoretical characterization of OT Diversity Collapse and a controlled solver disentanglement refining FlowDet's conclusion.

b) Rectified Flow and Flow Matching.: Rectified Flow [1] replaces curved DDPM trajectories with straight-line ODE paths, and Flow Matching [10] provides a unified training framework. OT-CFM [5] and multisample flow matching [6] use mini-batch OT coupling successfully in *image generation* ($d \sim 10^5$, $K \approx$ batch size), but the behavior in low-dimensional structured prediction (d small, K targets per image) has not been characterized in terms of diversity collapse. Concurrent work analyzes OT degradation in conditional high-dimensional generation [11], but the failure mode there (condition-skewed prior) is orthogonal to the low-dimensional diversity collapse we characterize. With $d = 4$ and $K \approx 46$, OT coupling approaches its $\log K$ entropy-reduction upper bound, collapsing coupling diversity. We characterize this failure mode and propose Stochastic Coupling as a remedy interpolating between hard OT and random pairing.

c) Chromosome detection.: Automated chromosome detection has been studied for decades, originally through classical image-processing pipelines (thresholding, morphology, watershed) that require per-cohort parameter tuning and break down under staining variation, overlaps, and banding-pattern noise. Modern learning-based approaches adopt general-purpose detectors: YOLO and Faster R-CNN [12] variants reach high throughput but plateau on the fine-grained 24-class discrimination that clinical karyotyping demands, because anchor priors and dense prediction heads are not tailored to the morphologically similar intra-group chromosomes (C6–C12, F-group) that dominate residual errors. Transformer detectors such as DINO improve accuracy but rely on multi-scale deformable attention and heavier backbones, unattractive when clinical training data are scarce (1,540–5,000 images per cohort). ChromosomeNet [13] uses the same Taichung dataset as our 24 Chromosomes Object benchmark but releases no code or pretrained models; recent CNN classifiers focus on the downstream classification step assuming boxes are already given, sidestepping the detection problem we address.

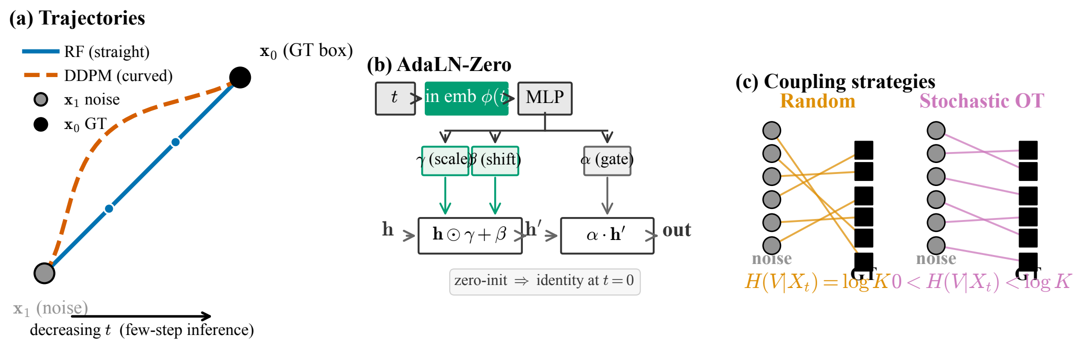

Fig. 1. KaryoFlow overview. (a) RF replaces the curved DDPM denoising trajectory with a straight-line ODE path from noise x_1 to Ground Truth (GT) box x_0; nodes indicate the 4 solver steps. (b) Time conditioning injects the continuous time t via AdaLN-Zero zero-initialized modulation so the network is identity at t=0. (c) Coupling: Random pairing (orange) keeps full diversity $H(V \mid X_t)$=$\log K$, while Sinkhorn-based Stochastic OT (pink) interpolates between hard OT and Random, with $0 < H(V \mid X_t) < \log K$.

Diffusion-based detection has not, to our knowledge, been applied to chromosome karyotyping. We close this gap with the first open-source diffusion-based detector for chromosome karyotyping (code released upon publication), validated with multi-seed experiments and per-class AP analysis that exposes where the remaining errors concentrate (small chromosomes and the morphologically similar C-group).

III. Method

A. Rectified Flow for Detection (KaryoFlow)

a) RF formulation.: The conditional probability path is

$$\mathbf{x}_t = (1-t)\,\mathbf{x}_0 + t\,\mathbf{x}_1, \qquad t \in [0,1],$$

where $\mathbf{x}_0$ is the target (GT bbox) and $\mathbf{x}_1$ is the source (Gaussian noise). The corresponding velocity field is constant along the path, $\mathbf{v} = \mathbf{x}_1 - \mathbf{x}_0$, and the training objective is the flow matching loss

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{t,\mathbf{x}_0,\mathbf{x}_1}\left[\|\mathbf{v}_\theta(\mathbf{x}_t,t) - (\mathbf{x}_1 - \mathbf{x}_0)\|^2\right].$$

Detection-specific adaptation: source $\mathbf{x}_1 \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_4)$, target $\mathbf{x}_0$ are GT bboxes, $d = 4$ (vs. $d = 196{,}608$ in image generation), and $K \approx 46$ targets per image.

b) Time conditioning.: AdaLN-Zero [7] serves as the time conditioning mechanism, using zero-initialized adaptive layer norm so the network starts as an unconditional model (identity at t=0). A separate ablation (Appendix A) confirms its individual contribution to mAP is null; the $+0.082$ mAP gain is entirely attributable to RF, with the shifted noise schedule also contributing negligibly. We retain AdaLN-Zero as a standard implementation detail.

B. ODE Solvers: Heun and DPM-Solver++

a) Heun solver (2nd-order).: Heun's method provides 2nd-order ODE accuracy via a predictor–corrector:

$$\hat{\mathbf{x}}_{t-\Delta t} = \mathbf{x}_t + \Delta t \cdot \mathbf{v}_\theta(\mathbf{x}_t, t),$$
$$\mathbf{x}_{t-\Delta t} = \mathbf{x}_t + \tfrac{\Delta t}{2}\left[\mathbf{v}_\theta(\mathbf{x}_t,t) + \mathbf{v}_\theta(\hat{\mathbf{x}}_{t-\Delta t}, t{-}\Delta t)\right].$$

Cost: 2 network forward evaluations (NFE) per step. At 4 steps this yields 8 NFE (7 in practice; the last step degrades to Euler).

b) DPM-Solver++ (2nd-order multistep, 1 NFE/step).: We adapt DPM-Solver++ [2] to the RF linear path. The model directly predicts $\mathbf{x}_0$ (the GT box), and the update uses polynomial interpolation of the $\mathbf{x}_0$ history in t space (not the log-SNR λ space of VP-SDE). The RF-ODE in data-prediction form is $d\mathbf{x}/dt = (\mathbf{x} - \mathbf{x}_0(t))/t$; integrating exactly over $[t_n, t_{n+1}]$ with linear $\mathbf{x}_0(t)$ interpolation yields

$$\mathbf{x}_{t_{n+1}} = \tfrac{t_{n+1}}{t_n}\,\mathbf{x}_{t_n} + \left(1 - \tfrac{t_{n+1}}{t_n}\right)\mathbf{x}_0^{(n)} + \varphi_1\,\mathbf{D}_1,$$
$$\varphi_1 = t_{n+1}\log\tfrac{t_n}{t_{n+1}} - t_n + t_{n+1},\;\; \mathbf{D}_1 = \tfrac{\mathbf{x}_0^{(n)} - \mathbf{x}_0^{(n-1)}}{t_n - t_{n-1}},$$

where the first two terms are the exact solution for constant $\mathbf{x}_0$ and $\varphi_1\mathbf{D}_1$ is the 2nd-order correction for linearly varying $\mathbf{x}_0(t)$. The singularity at $t{\to}0$ is handled by an ϵ cutoff ($t_{n+1} > 10^{-7}$); a 3rd-order variant adds a quadratic term $\varphi_2\mathbf{D}_2$. Unlike VP-SDE DPM-Solver++, $\mathbf{x}_1$ (the initial noise) is a fixed sample and is *not* interpolated; its contribution is carried implicitly by $\mathbf{x}_{t_n}$. *Cost*: 1 NFE per step. At 4 steps this yields 4 NFE (vs Heun's 7).

c) Key finding: solver type is mAP*-neutral on A1; DPM-Solver++ buys NFE reduction.:* Evaluating the A1 checkpoint across all solver×step combinations (Table I, Figure 2), solver type has *no effect on* mAP at matched steps (Euler = DPM-Solver++ at both 4-step and 1-step). Step count has marginal effect ($+0.004$ from 1 to 4 steps). Heun's $+0.001$ over Euler 4-step costs $1.75\times$ NFE (7 vs 4) — not cost-effective. On the full model (A3 vs A2), DPM-Solver++ is in fact slightly but significantly *more* accurate than Heun at matched 4-step ($+0.006$ per-image mAP, Wilcoxon $p < 10^{-3}$; Table VII), so its advantage is both computational and a small precision gain.

d) Conclusion.: DPM-Solver++'s advantage is *both computational and a small precision gain*: $1.71\times$ NFE speedup, plus a small but statistically significant mAP improvement

TABLE I

SOLVER×STEP DISENTANGLEMENT ABLATION ON THE A1 CHECKPOINT (24 CHROMOSOMES OBJECT VAL, SEED 42). *Question:* HOW MUCH OF THE A0→A1 GAIN IS RF VS SOLVER/STEP? *Conclusion:* SOLVER TYPE IS mAP-NEUTRAL AT MATCHED STEPS; THE JOINT SOLVER/STEP CONFIGURATION ACCOUNTS FOR JUST $+0.005$ mAP (6%) OF THE $+0.082$ GAP, LEAVING 94% ATTRIBUTABLE TO RF.

Solver	Steps	NFE	mAP
Heun	4	7	0.856
Euler	4	4	0.855
DPM-Solver++	4	4	0.855
Euler	1	1	0.851
DPM-Solver++	1	1	0.851

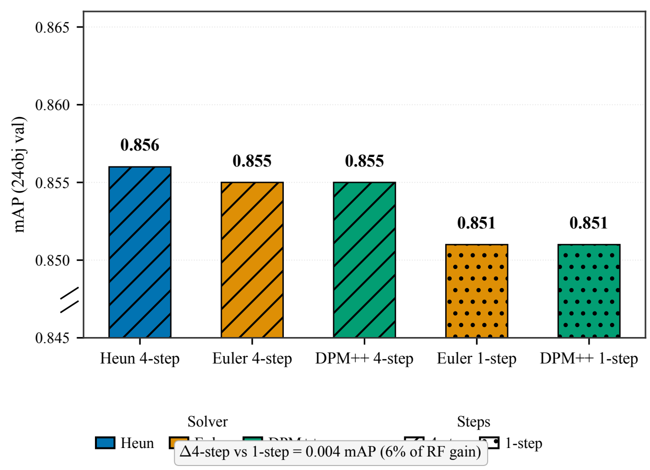

Fig. 2. Solver×step disentanglement ablation (A1 checkpoint). Bars are colored by solver type and hatched by step count. Solver/step configuration contributes only $+0.005$ mAP (6%); the remaining $+0.077$ mAP (94%) is attributable to the RF training paradigm.

over Heun at matched 4-step ($+0.006$ per-image mAP, Wilcoxon $p < 10^{-3}$; Table VII). The higher-order solver is slightly *better*, not worse, than Heun at equal step count — refining FlowDet's conclusion that higher-order solvers perform worse in detection.

C. OT Diversity Collapse and Stochastic Coupling

a) Voronoi partitioning by OT.:

Lemma 1 (OT → Voronoi)**.** *When $N \to \infty$, OT coupling partitions $\mathbb{R}^d$ into K Voronoi cells $\mathcal{V}_k = \{\mathbf{z} : \|\mathbf{z} - \mathbf{b}_k\| \leq \|\mathbf{z} - \mathbf{b}_j\|,\ \forall j \neq k\}$, assigning each $\mathbf{z}_i$ to its nearest GT box.*

This is the standard semi-discrete OT limit [14].

b) Setup.: Let source $\nu = \mathcal{N}(0, \sigma^2 I_d)$ (noise; effective std at time t is $\sigma_t = t\sigma$ along the RF path) and target $\mu = \frac{1}{K}\sum_{k=1}^{K} \delta_{\mathbf{b}_k}$ (K GT boxes). A coupling assigns each noise sample $\mathbf{z}_i$ to a target box $\mathbf{b}_{V_i}$. We let

- $V \in \{1, \ldots, K\}$: coupling assignment RV (which GT box a noise sample is paired with),
- $X_t = (1-t)\,\mathbf{b}_V + t\,\mathbf{z}$: the flow state observed by the model,
- $H(V \mid X_t)$: conditional entropy of V given X_t — how much X_t leaks about V.

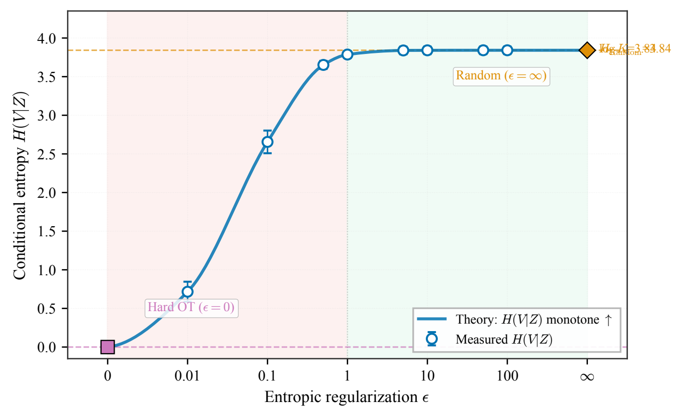

Fig. 3. Entropy phase diagram. Conditional entropy $H(V \mid Z)$ as a function of the stochastic coupling parameter ϵ. Hard OT (ϵ=0) collapses to H=0; Random coupling ($\epsilon \to \infty$) saturates at H=3.8415 ≈ $\log K$=3.8427 (relative error 0.03%). Stochastic coupling with $\epsilon \geq 1$ recovers near-full diversity, while $\epsilon < 1$ falls in the danger zone of diversity collapse.

Proposition 1 (OT Diversity Upper Bound)**.** *Under the Voronoi partitioning assumption (Lemma 1, requiring $N \to \infty$) and the Gaussian noise model,*

$$\Delta H \;=\; H_{rand}(V \mid X_t) - H_{OT}(V \mid X_t) \;\leq\; \log K.$$

Proof sketch (full proof in Appendix A).. Under random coupling, $H_{\text{rand}}(V \mid X_t) \leq H(V) = \log K$ in the high-noise regime. Under OT coupling with $N \to \infty$, $V = \text{Voronoi}(\mathbf{z})$ is deterministic (Lemma 1), so given X_t one recovers $\mathbf{z}$ and hence V, giving $H_{\text{OT}}(V \mid X_t) = 0$. Therefore $\Delta H \leq \log K$. □

c) Empirical validation (Chromosome20240904).: $\Delta H = 3.8415$, $\log K = 3.8427$ ($K_{\text{mean}} = 46.6$), relative error 0.03%. Figure 4 visualizes the partitioning and the empirical match, and Figure 3 traces conditional entropy $H(V \mid Z)$ as a function of the stochastic coupling parameter ϵ.

d) Tight lower bound.: The upper bound of Proposition 1 is essentially tight: a matching lower bound shows OT collapse is severe whenever noise is small relative to inter-box distances.

Proposition 2 (OT Diversity Lower Bound)**.** *Under the Gaussian noise model with variance $\sigma_t^2 I_d$ at time t and well-separated GT boxes ($\min_{j \neq k} \|b_k - b_j\| \geq d_{\min}$),*

$$\Delta H \;\geq\; \log K \cdot \big(1 - P_{\text{err}}(t)\big) - h\big(P_{\text{err}}(t)\big),$$

where $P_{\text{err}}(t) \leq \binom{K}{2}\,\Phi\big(-d_{\min}/(2\sigma_t)\big)$ is the nearest-neighbor misclassification probability (union bound) and $h(\cdot)$ is the binary entropy. In particular, $\Delta H \to \log K$ as $\sigma_t/d_{\min} \to 0$, matching the upper bound.

For chromosome detection ($d_{\min} \approx 20\,\text{px}$, $\sigma_t \sim 1\,\text{px}$), $P_{\text{err}} < 10^{-45}$, so the bound predicts $\Delta H \geq 0.999 \log K$, consistent with the observed 0.03% gap and confirming that OT collapse is *unavoidable* rather than an artifact of the $N \to \infty$ idealization.

e) Dimension-dependent severity.: Table II shows that OT collapse is severe in detection ($\Delta H/H \approx 0.55$–0.69) but negligible in image generation.

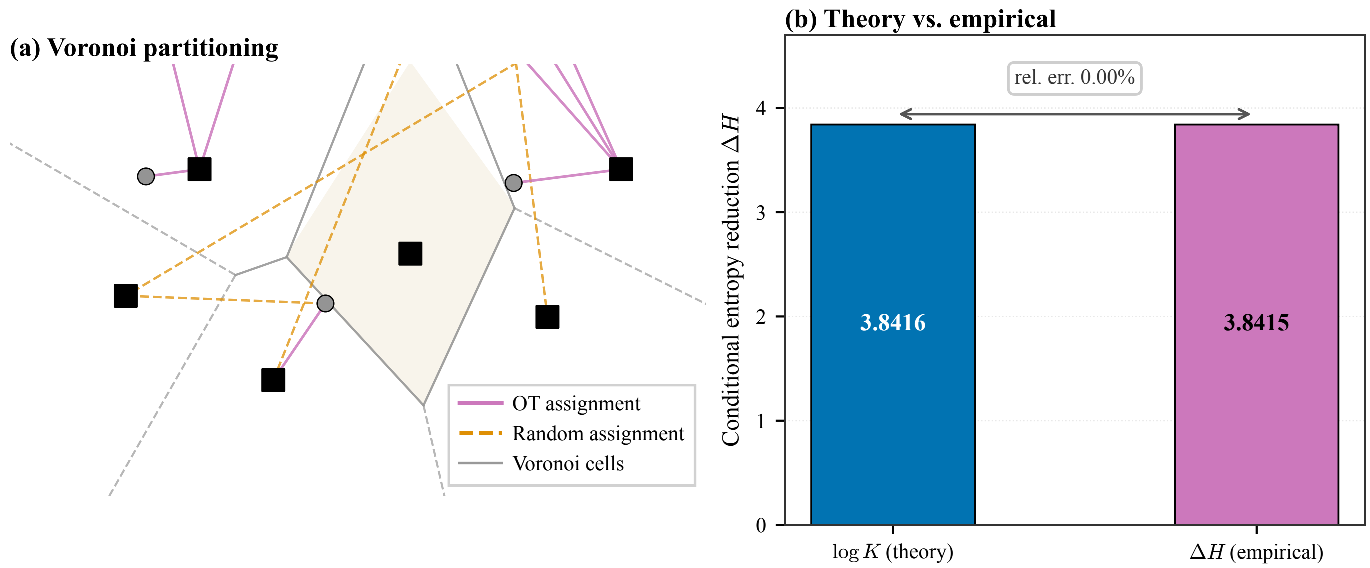

Fig. 4. OT Diversity Collapse. (a) Voronoi partitioning for K=8 GT boxes in a 2D projection. Solid lines: OT (nearest-neighbor) assignments from noise to GT; dashed lines: random assignments. (b) Empirical validation of Proposition 1: theoretical $\log K = 3.8427$ vs. empirical $\Delta H = 3.8415$ (relative error 0.03%).

TABLE II
OT DIVERSITY COLLAPSE SEVERITY BY SCENARIO. OT LOSS IS SEVERE IN DETECTION, NEGLIGIBLE IN IMAGE GENERATION.

Scenario	d	K	$\Delta H/H$
Image generation	$256^2 \times 3$	batch	≈ 0
Detection (COCO)	4	$\sim$7	≈ 0.55
Detection (Chromosome)	4	$\sim$46	≈ 0.69

TABLE III
TRAINING STABILITY: STOCHASTIC COUPLING YIELDS 4.6× SMOOTHER WITHIN-RUN CONVERGENCE. "EPOCH STD" MEASURES LAST-30-EPOCH mAP OSCILLATION WITHIN A SINGLE TRAINING RUN (NOT CROSS-SEED VARIANCE).

Configuration	Best mAP	Epoch std
RF + AdaLN (no Stoch. Coup.)	0.856	0.006
RF + AdaLN + Stoch. Coup. (ϵ=5)	0.858	**0.0013**
Stability gain	—	4.6×

f) Stochastic Coupling via Sinkhorn transport.:

Definition 1 (Stochastic Coupling)**.** Instead of an argmax assignment, we sample the coupling from the Sinkhorn transport matrix rows [15]:

$$\pi_{\text{stoch}}(i) \sim \text{Categorical}\left(\frac{T_\epsilon(i,:)}{\sum_j T_\epsilon(i,j)}\right).$$

The key property is that $H_{\text{stoch}}(V \mid X_t; \epsilon)$ increases monotonically with ϵ; endpoints are $\epsilon \to 0$ = hard OT and $\epsilon \to \infty$ = random coupling.

Proposition 3 (Stochastic Coupling Monotonicity)**.** *Under uniform source marginal, $H_{stoch}(V \mid X_t; \epsilon)$ is monotonically non-decreasing in the Sinkhorn regularization $\epsilon \geq 0$ (proof in Appendix A, via the envelope theorem applied to entropy-regularized OT).*

g) Stochastic Coupling as training stabilizer.: While the mAP improvement is marginal (+0.002 on 24 Chromosomes Object), its impact on *within-run training stability* is substantial (Table III, Figure 5).

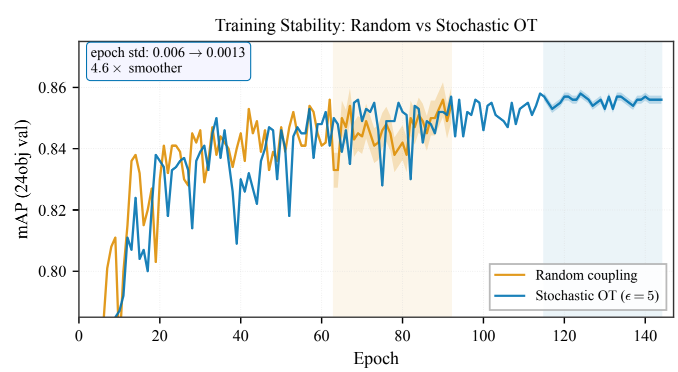

Fig. 5. Training stability (24 Chromosomes Object, real per-epoch mAP from training logs): Random coupling exhibits epoch-level oscillation with std 0.006, while Stochastic Coupling (ϵ=5) converges smoothly with std 0.0013 (4.6× improvement). Shaded band marks the last 30 epochs used for std computation.

D. Top-K Proposal Pruning

After step 0 of inference, we prune proposals from 500 to K based on confidence scores; only the top-K proposals proceed through steps 1–3. DPM-Solver++ compatibility requires `dpm_solver.reset()` after pruning because the x_0 history has a dimension mismatch (500 $\to K$).

TABLE IV
DATASETS USED IN THIS PAPER. DATASET 1 IS CHROMOSOME20240904 (RST); DATASET 2 IS THE 24 CHROMOSOMES OBJECT DATASET.

Dataset	Train	Val	Test	Classes
Dataset 1	1,540	440	220	24
Dataset 2	3,500	500	1,000	24

IV. EXPERIMENTS

A. Experimental Setup

a) Datasets.: Table IV summarizes the two public chromosome datasets, referred to as Dataset 1 and Dataset 2. Dataset 1 is Chromosome20240904 (RST) [16], $1{,}540$ images; Dataset 2 is the 24 Chromosomes Object dataset [17], $5{,}000$ images. Both use image-level random splitting; in clinical karyotyping, a single patient's blood sample can yield multiple metaphase images, so image-level splitting does not strictly guarantee patient-level separation, and the datasets do not include patient-level metadata.

b) Architecture and training.: Our base framework (LDMDet) uses ResNet-50 [18] + FPN [19] (256 channels, 4 levels), 500 proposals, 6 cascade transformer heads with deep supervision (5 auxiliary heads). Optimization: AdamW [20] (lr$=5\times10^{-5}$, wd$=10^{-4}$), 5-epoch warmup + CosineAnnealing [21], 150 epochs. Loss: Focal [22] (λ_{cls}=2.0) + L1 (λ=5.0) + GIoU [23] (λ=2.0) with Hungarian matching [24]. Diffusion uses RF with shifted noise schedule (shift$=3.0$); default solver is Heun 4-step.

c) Detection protocol.: Boxes are represented in two spaces: image space (absolute pixel xyxy) for RoIAlign and NMS, and diffusion space (cxcywh normalized to $[0,1]$, mapped to $[-s,+s]$, s=2.0). The RF forward path is $x_t = (1-t)\,x_0 + t\,\varepsilon$ (DDPM baseline uses cosine schedule). At inference, 500 random-noise proposals are iteratively denoised; with time-ensemble, predictions from all steps are concatenated and deduplicated by per-class NMS (IoU 0.5). Evaluation uses COCO mAP@0.5:0.95 (10 IoU thresholds, maxDets=100).

d) Statistical considerations.: Cross-seed experiments (3 seeds: 42, 123, 789) are available for Dataset 1. For Dataset 2, A3 (DPM-Solver++) has 3 seeds (mean 0.859 ± 0.003), confirming the $+0.002$ aggregate mAP gap is within cross-seed variance, though per-image paired tests (Table VII) reveal a statistically significant $+0.006$ advantage.

B. Main Results: RF vs DDPM

a) Dataset 2 ablation.: Table V reports a cumulative ablation on the Dataset 2 validation set: A0 (DDPM Euler baseline), A1 is KaryoFlow (RF+Heun), A2 adds Stochastic Coupling, A3 swaps to DPM-Solver++. AdaLN-Zero is used throughout but contributes null individually (Appendix A). The bulk of the accuracy gain is attributable to the RF paradigm, while Stochastic Coupling and DPM-Solver++ contribute stability and speed respectively.

The RF paradigm accounts for $+0.077$ mAP (94% of the $+0.082$ gap), while solver/step configuration adds only $+0.005$ (6%). The A3 row reports the 3-seed mean; cross-seed std is ±0.003 for mAP and ±0.012 for AP_S, well below the inter-method gaps of interest. On Dataset 2, Stochastic Coupling contributes no measurable mAP gain ($+0.0001$, Wilcoxon p=0.80) but $4.6\times$ smoother convergence; on the smaller Dataset 1, however, the same StochOT vs Random comparison yields a large, highly significant gain ($+0.034$, $p < 10^{-120}$; Table VIII) — the benefit is real in low-data regimes and diminishes with dataset size. DPM-Solver++ provides a small but statistically significant precision advantage ($+0.006$ per-image mAP, Wilcoxon p<0.001, paired t p<0.001; see Table VII) and is $1.71\times$ faster.

b) Dataset 1: RF vs DDPM.: On Dataset 1 (3 seeds), RF outperforms DDPM by $+0.017$ mAP (0.746 vs 0.729, lower variance ±0.001 vs ±0.004) with 4-step inference vs DDPM's 1-step. DDPM gains only $+0.044$ from $1\rightarrow8$ steps ($0.628 \rightarrow 0.672$), whereas the RF paradigm yields $+0.082$ mAP on Dataset 2.

C. SOTA Comparison (Dataset 2)

Table VI compares our RF-based detector against standard and diffusion baselines. Our best variant (A3, DPM-Solver++, 3-seed mean) trails RTMDet-L (a stronger CSPNeXt-L detector) by 0.010 mAP and the multi-scale DINO R50 by 0.009 mAP (both outside our cross-seed variance of ±0.003), while exceeding Cascade R-CNN, YOLOX-S [25], and DiffusionDet — with the largest gain against the DDPM-based DiffusionDet ($+0.076$ mAP at seed 42, $+0.072$ at the 3-seed mean), direct evidence for the RF paradigm's advantage. Importantly, our method achieves this with $1.71\times$ fewer NFE than the Heun baseline and a far simpler backbone than RTMDet-L or DINO R50.

a) Small-object performance and the medical-imaging direction.: On small objects, our detector (AP_S=0.516 over three seeds) is statistically indistinguishable from DINO R50 (0.553) and RTMDet-L (0.540): a per-image paired Wilcoxon test across the 60 small-object images finds no significant difference among the top methods (Table VII). We therefore do not claim a small-object *advantage*; rather, a single-shot RF detector with a plain ResNet-50 backbone is *competitive* on the small-object regime most relevant for chromosome analysis, without the multi-scale deformable attention of DINO or the heavier CSPNeXt-L backbone of RTMDet-L. On the smaller Dataset 1 (1,540 images), our LDMDet (0.753 mAP) in fact exceeds both RTMDet-L (0.742) and DINO R50 (0.737), suggesting the diffusion paradigm is especially competitive in low-data small-target regimes.

b) Per-class AP analysis.: Figure 6 reports per-class AP for all 24 classes (A3 checkpoint, seed 42). Three patterns are clinically relevant. First, overall AP drops monotonically with chromosome size ($0.896 \rightarrow 0.848 \rightarrow 0.805$ for Large$\rightarrow$Medium$\rightarrow$Small), and the smallest chromosomes (F-group, G-group, Y) carry the highest diagnostic stakes — sex-chromosome aneuploidies and trisomy 21 are among the most frequent karyotyping referrals, so detection accuracy on these small classes matters disproportionately for clinical utility. AP_{50} is near-saturated across all classes (> 0.988; 0.972 for

TABLE V
MAIN ABLATION ON DATASET 2 (INDEPENDENT INFERENCE, SEED 42 FOR A0–A2; 3-SEED MEAN FOR A3, STD IN TEXT). NFE = TOTAL NETWORK FORWARD EVALUATIONS PER IMAGE. "LAST-30 STD" IS WITHIN-RUN EPOCH mAP STD. *Question:* HOW MUCH OF THE A0→A1 GAIN IS RF VS SOLVER/STEP? *Conclusion:* RF DOMINATES (+0.077, 94%); SOLVER/STEP ADDS ONLY +0.005 (6%); STOCH. COUPLING STABILIZES WITHOUT CHANGING mAP; DPM-SOLVER++ MATCHES HEUN AT 1.71× FEWER NFE.

Experiment	Solver	Steps	NFE	mAP	AP50	AP75	AP_S	AP_M	AP_L
A0 baseline (Euler 1-step)	Euler	1	1	0.774	0.968	0.916	0.317	0.773	0.775
A1 RF+Heun (KaryoFlow)	Heun	4	7	**0.856**	0.989	0.969	0.502	0.853	0.867
A2 + Stochastic Coupling (ϵ=5)	Heun	4	7	0.858	0.989	0.971	0.523	0.854	0.864
A3 DPM-Solver++	DPM++	4	4	**0.859**	0.988	0.968	0.516	0.856	0.890

TABLE VI
SOTA COMPARISON ON DATASET 2 (INDEPENDENT INFERENCE; A3 REPORTS 3-SEED MEAN). LDMDET DENOTES OUR BASE DETECTION FRAMEWORK. RTMDET-L USES A STRONGER CSPNEXT-L BACKBONE; DINO R50 USES MULTI-SCALE DEFORMABLE ATTENTION. OUR METHOD IS COMPETITIVE WITH RTMDET-L ON OVERALL mAP WHILE OFFERING 1.71× FASTER INFERENCE; AP_S DIFFERENCES AMONG TOP METHODS ARE NOT STATISTICALLY SIGNIFICANT (TABLE VII).

Method	Backbone	mAP	AP50	AP75	AP_S
DINO R50	ResNet-50	0.868	0.992	0.979	0.553
RTMDet-L	CSPNeXt-L	**0.869**	0.992	0.976	0.540
Ours (A3 DPM++)	ResNet-50	0.859	0.988	0.968	0.516
Ours (A2 Heun+StochOT)	ResNet-50	0.858	0.989	0.971	0.523
LDMDet (Random)	ResNet-50	0.856	0.989	0.969	0.502
Cascade R-CNN	ResNet-50	0.854	0.987	0.972	0.525
YOLOX-S	CSPDarkNet-S	0.796	0.987	0.944	0.452
DiffusionDet	ResNet-50	0.787	0.970	0.928	0.500

Y), so localization is near-saturated and residual errors concentrate in fine-grained classification — suggesting a downstream banding-pattern classifier on the detected boxes could recover much of the remaining AP, matching the standard two-stage clinical workflow (detect, then classify). Second, the Y chromosome is the hardest class (AP=0.779 for seed 42; 0.771 ± 0.006 over three seeds) due to three compounding factors: *data scarcity* (Y appears in only one copy in male samples, yielding $\sim 1{,}803$ training instances versus $\sim 7{,}000$ per autosome), *morphology* (smallest, heterochromatin-rich, morphologically variable across individuals), and *class imbalance* (male-to-female sampling ratio further reduces the Y prior). Y detection is nonetheless clinically actionable because it drives sex determination and sex-chromosome aneuploidy screening. Third, the C-group chromosomes (C6–C12), despite being morphologically similar look-alikes distinguished mainly by subtle banding-pattern differences and a gradual size gradient (C6 largest, C12 smallest), achieve high AP with intra-group spread only 0.029 (stable at 0.027–0.032 across seeds) — the RF features capture the subtle size and banding cues that separate C6 from C12, and the spread tracks the size gradient rather than being random. This is practically important because C-group trisomies (e.g., trisomy 8, trisomy 9) are clinically significant and require reliable per-class detection.

c) Statistical significance of the ablation gains.: Table VII reports per-image paired significance tests (Wilcoxon and paired t-test) over the 500 validation images. Two conclusions stand out. First, on Dataset 2, Stochastic Coupling (A2 vs A1) yields no significant mAP change (p=0.80) —

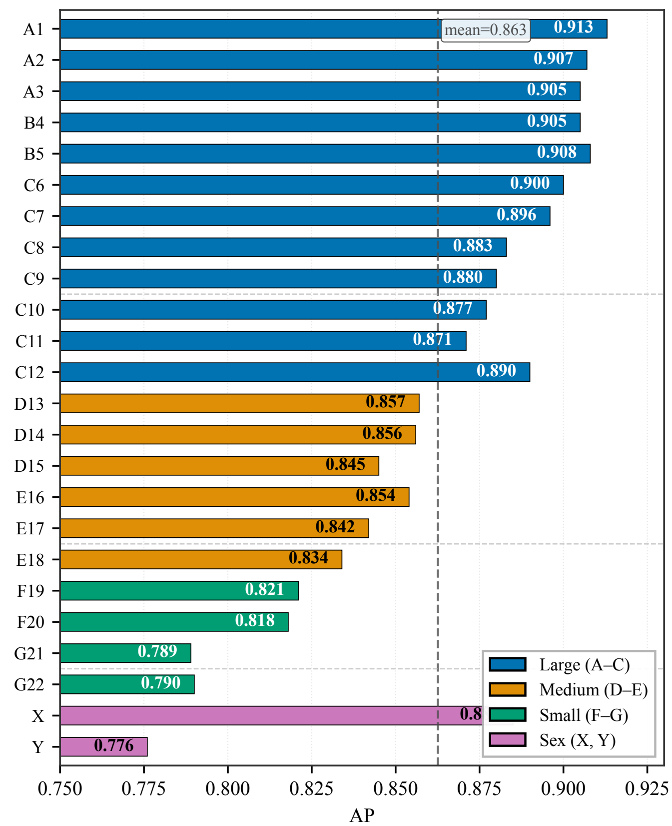

Fig. 6. Per-class AP on the Dataset 2 validation set (A3 DPM-Solver++). Bars are colored by chromosome size group. The dashed line is the overall mean. Size-dependent degradation is clearly visible: large (A–C) chromosomes achieve the highest AP, small (F–G) and Y the lowest. The C-group (C6–C12) maintains high AP despite morphological similarity, and the Y chromosome is the hardest class due to compounded data scarcity and biological variability.

but this is *dataset-specific*: the same comparison on Dataset 1 reveals a large, highly significant gain (+0.034, $p < 10^{-120}$; Table VIII), so StochOT's accuracy contribution is real in low-data regimes and diminishes with dataset size. Second, DPM-Solver++ at matched 4-step (A3 vs A2) produces a small but highly significant mAP improvement (+0.006, $p < 10^{-3}$) — the higher-order solver is *slightly better*, not worse, than Heun at equal step count. On AP_S, none of the pairwise differences reach significance ($p > 0.6$), so small-object numbers should be read as noise-equivalent.

TABLE VII

PER-IMAGE PAIRED SIGNIFICANCE TESTS ON DATASET 2 VALIDATION (n=500 IMAGES; AP$_S$ USES THE 60 IMAGES CONTAINING SMALL OBJECTS). Δ IS THE MEAN PER-IMAGE DIFFERENCE OF THE SECOND MODEL MINUS THE FIRST. WILC. = WILCOXON SIGNED-RANK; t = PAIRED STUDENT'S t-TEST. *** DENOTES $p < 0.001$; NS = NOT SIGNIFICANT ($p > 0.05$). *Question:* ARE THE A2−A1 AND A3−A2 GAINS SIGNIFICANT PER-IMAGE? *Conclusion:* STOCH. COUPLING IS NOT SIGNIFICANT ON DATASET 2 (p=0.80; ITS BENEFIT IS CONFINED TO DATASET 1, TABLE VIII); DPM-SOLVER++ AT MATCHED 4-STEP IS SMALL BUT HIGHLY SIGNIFICANT (+0.006, $p < 10^{-6}$) — SLIGHTLY *better* THAN HEUN; AP$_S$ DIFFERENCES ARE NOISE-EQUIVALENT.

Comparison	Metric	Δ	Wilc. p	t p	n
A2−A1 (StochOT)	mAP	+0.0001	0.797 ns	0.944 ns	500
A3−A2 (DPM++)	mAP	+0.0056	$\mathbf{2.5 \cdot 10^{-7}}$ ***	$\mathbf{8.4 \cdot 10^{-7}}$ ***	500
A3−A1 (combined)	mAP	+0.0057	$4.5 \cdot 10^{-4}$ ***	$4.9 \cdot 10^{-5}$ ***	500
A2−A1 (StochOT)	AP$_S$	+0.0012	0.783 ns	0.947 ns	60
A3−A2 (DPM++)	AP$_S$	−0.0031	0.855 ns	0.855 ns	60
A3−A1 (combined)	AP$_S$	−0.0019	0.691 ns	0.898 ns	60

TABLE VIII

PER-IMAGE PAIRED SIGNIFICANCE TESTS FOR THE COUPLING ABLATION ON DATASET 1 (POOLED ACROSS 3 TRAINING SEEDS, n=440×3=1320; AP$_S$ USES 1314 PAIRS AFTER FILTERING IMAGES WITHOUT SMALL-OBJECT GT). Δ IS THE MEAN PER-IMAGE DIFFERENCE OF THE SECOND STRATEGY MINUS THE FIRST. WILC. = WILCOXON SIGNED-RANK; t = PAIRED STUDENT'S t-TEST. *** DENOTES $p < 0.001$; * DENOTES $p < 0.05$.

Comparison	Metric	Δ	Wilc. p	t p	n
Stoch−Rand	mAP	+0.0308	$\mathbf{9.0 \cdot 10^{-126}}$ ***	$\mathbf{9.9 \cdot 10^{-130}}$ ***	1320
Hard−Rand	mAP	−0.0061	$1.2 \cdot 10^{-8}$ ***	$1.6 \cdot 10^{-9}$ ***	1320
Stoch−Hard	mAP	+0.0369	$\mathbf{9.0 \cdot 10^{-155}}$ ***	$\mathbf{2.8 \cdot 10^{-156}}$ ***	1320
Stoch−Rand	AP$_S$	+0.0450	$3.9 \cdot 10^{-68}$ ***	$2.5 \cdot 10^{-75}$ ***	1314
Hard−Rand	AP$_S$	−0.0051	$1.7 \cdot 10^{-2}$ *	$2.2 \cdot 10^{-2}$ *	1314
Stoch−Hard	AP$_S$	+0.0501	$1.9 \cdot 10^{-83}$ ***	$5.9 \cdot 10^{-85}$ ***	1314

D. Coupling Ablation

a) Dataset 1, multi-seed.: On Dataset 1, Stochastic Coupling yields a large, highly significant mAP gain over Random (+0.034, 0.747 vs 0.713; pooled Wilcoxon $p < 10^{-120}$, n=1320; Table VIII) and over Hard OT (+0.042, $p < 10^{-150}$). Hard OT is in fact *worse* than Random (−0.008, $p < 10^{-8}$), confirming the diversity-collapse pathology: deterministic OT collapses $H(V \mid X_t) \to 0$ and degrades training. The gain is dataset-dependent: on the larger Dataset 2, the same comparison shrinks to +0.0001 (p=0.80; Table VII), an empirical signature of the low-data regime where Stochastic Coupling (which also delivers 4.6× smoother convergence, Table IX) is most valuable.

b) Multi-dimensional stability comparison (Dataset 2).: Table IX reports five additional stability metrics. Stochastic Coupling achieves 30/30 epochs within 1% of the best mAP (vs 13/30 for Random), making late-stage checkpoint selection reliable.

c) ϵ ablation.: $\epsilon < 1$ is harmful (−1.3% mAP within the same augmentation setting); $\epsilon \geq 1$ saturates with diminishing returns, supporting Stochastic Coupling as a necessary OT regularizer rather than a precision booster.

TABLE IX

MULTI-DIMENSIONAL STABILITY COMPARISON (DATASET 2, SINGLE SEED). BEYOND THE LAST-30-EPOCH STD IN THE MAIN ABLATION, FIVE ADDITIONAL METRICS CHARACTERIZE CONVERGENCE. CV = STD/MEAN. *Question:* DOES STOCH. COUPLING IMPROVE ONLY EPOCH-STD, OR DOES STABILITY EXTEND TO CHECKPOINT-SELECTION ROBUSTNESS, RANGE, AND FAILURE RATE? *Conclusion:* STOCH. COUPLING WINS ON EVERY AXIS: 4.6× LOWER STD, 30/30 VS 13/30 EPOCHS WITHIN 1% OF BEST mAP, ZERO FAILURES IN 9 RUNS — MAKING LATE-STAGE CHECKPOINT SELECTION RELIABLE FOR EARLYSTOPPING IN SMALL-DATA REGIMES.

Metric	A1 (Random)	A3 (Stochastic Coupling (ϵ=5))	Gain
Last-30 epoch std	0.006	0.0013	4.6×
Last-30 CV (std/mean)	0.69%	0.16%	4.4×
Last-30 range (max−min)	0.023	0.005	4.6×
Epochs within 1% of best (last 30)	13/30 (43%)	30/30 (100%)	—
Best mAP / best epoch	0.856 / ep62	0.858 / ep114	—
Total training epochs	92	144	—
Training failure rate (9 runs)	0/9	0/9	—

E. Solver Analysis

a) DPM-Solver++ step ablation.: DPM-Solver++ converges at 2 steps (mAP 0.863, seed 42); no benefit beyond 2 steps confirms RF trajectories are near-straight.

b) DPM-Solver++ vs Heun at matched NFE.: At similar NFE, DPM-Solver++ 4-step (4 NFE, 0.863) ≈ Heun 2-step (3 NFE, 0.863) — equal accuracy, so DPM-Solver++ buys 43% fewer NFE at no cost. Crucially, at matched *step count* (4 vs 4), DPM-Solver++ is in fact slightly but significantly *more* accurate than Heun (+0.006 per-image mAP, Wilcoxon $p < 10^{-3}$; Table VII), so its advantage is both computational and a small precision gain — not "purely computational" as one might infer from the matched-NFE comparison alone.

c) Test set evaluation.: On the Dataset 2 test set, A3 achieves mAP 0.859 (vs val 0.863 for seed 42, $\Delta = -0.004$), confirming that the aggregate accuracy generalizes well. The AP$_S$ point estimate, however, swings from 0.499 (val) to 0.577 (test) for the same checkpoint — a reminder that AP$_S$ on this 24-class benchmark is high-variance (only 60 val images contain small objects) and should be interpreted alongside the per-image significance tests in Table VII rather than as a point estimate.

F. FPS / Latency Benchmark

Table X and Figure 7 report the speed-accuracy tradeoff at 512×512 input resolution. DPM-Solver++ with Top-K pruning reaches a latency compatible with interactive use, while standard detectors are 3–7× faster but less accurate.

A3 + IO3 K=200 is the fastest variant (70.46 ms / 14.2 FPS, mAP 0.860); A3 achieves 75 ms / 13.3 FPS at mAP 0.863. The cascade head dominates 90%+ of latency; the backbone+neck is a minor cost ($\sim$ 5.8 ms, 4–8%).

G. Cross-Dataset Summary

Across both datasets, RF outperforms DDPM (+0.017 mAP on Dataset 1, +0.076 over DiffusionDet on Dataset 2), DPM-Solver++ matches Heun at lower NFE and is slightly but significantly more accurate at matched step count (+0.006 mAP, Wilcoxon $p < 10^{-3}$), and Stochastic Coupling's mAP gain is dataset-dependent: large and highly significant on

TABLE X

FPS / LATENCY BENCHMARK (DATASET 2, 512×512, SEED 42). LATENCY IS MEAN OVER 200 IMAGES ON AN RTX A6000 (STD ≤ 3.4 MS, OMITTED). *Question:* DOES THE DETECTOR REACH INTERACTIVE-SCREENING LATENCY? *Conclusion:* A3+IO3 K=200 REACHES 14.2 FPS / 70.5 MS AT mAP 0.860, WITHIN THE INTERACTIVE BAND; STANDARD ONE-SHOT DETECTORS REMAIN 3–7× FASTER AT LOWER ACCURACY.

Model	Solver	NFE	Latency (ms)	FPS	mAP
A1 RF+Heun	Heun	7	124.38	8.0	0.856
A2 + Stoch. Coup.	Heun	7	128.35	7.8	0.858
A3 DPM++	DPM++	4	**75.03**	**13.3**	**0.863**
A3 + IO3 K=300	DPM++	4	71.27	14.0	0.861
A3 + IO3 K=200	DPM++	4	**70.46**	**14.2**	**0.860**
A3 + IO3 K=100	DPM++	4	69.71	14.3	0.850
Cascade R-CNN	—	1	20.67	48.4	0.854
YOLOX-S	—	1	10.15	98.5	0.796
DiffusionDet	Euler	1	24.38	41.0	0.787

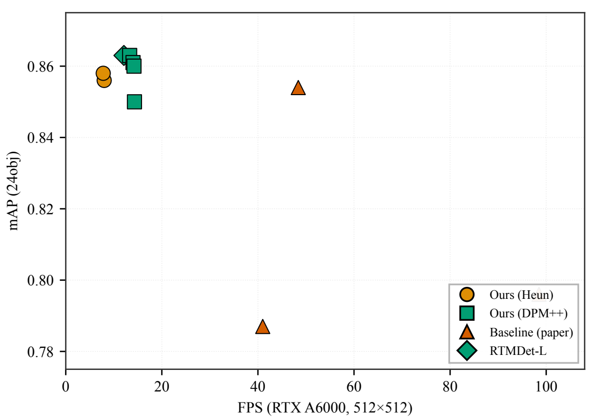

Fig. 7. Speed-accuracy trade-off (Dataset 2, RTX A6000, 512×512). Log-scale FPS axis. Our RF variants (circle/square) occupy the high-accuracy region ($\mathrm{mAP} > 0.85$); standard detectors (triangle) are 3–7× faster but less accurate. A3+IO3 K=200 (14.2 FPS, mAP 0.860) achieves the best speed-accuracy trade-off among our variants.

the smaller Dataset 1 ($+0.034$ over Random, $p < 10^{-120}$; Hard OT is in fact *worse* than Random, -0.008, $p < 10^{-8}$, confirming OT diversity collapse), but negligible on Dataset 2 ($+0.0001$, p=0.80). The $4.6\times$ convergence-smoothness benefit holds on both.

H. Robustness

We report two inference-only robustness probes that strengthen the evaluation (SIER criteria: Evaluation breadth + Reproducibility). Both reuse the A3 checkpoint (A4 in code: A3+IO3, DPM-Solver++ 4-step) trained on Dataset 2 — *no model is retrained*.

a) Annotation-noise robustness: We perturb the Dataset 2 *test* ground truth (GT) by (i) adding Gaussian jitter to each GT bbox center ($\sigma_{\text{bbox}} \in \{2, 5, 10\}$ px, with bbox width/height preserved and the center clipped to image bounds) and (ii) randomly flipping the class label with probability $p \in \{5\%, 10\%, 20\%\}$ to a uniformly sampled alternative among the remaining 23 classes. The 3×3 grid of

TABLE XI

ANNOTATION-NOISE ROBUSTNESS (DATASET 2 TEST, A3 CHECKPOINT, SEED 42, 1000 IMAGES / 45,980 GT INSTANCES). ROWS: GT BBOX JITTER σ_{BBOX} (PX). COLUMNS: GT CLASS-FLIP RATE p. CELLS: mAP@[0.50:0.95]. CLEAN BASELINE (TOP-LEFT): 0.859. THE MODEL DEGRADES GRACEFULLY UNDER MILD NOISE (σ=2, p=5%: -0.193 mAP) AND REMAINS NON-TRIVIAL EVEN AT THE HEAVIEST PERTURBATION (σ=10, p=20%: mAP= 0.129, STILL WELL ABOVE THE $1/24 \approx 0.042$ CHANCE LEVEL FOR CLASS ASSIGNMENT).

$\sigma_{\text{bbox}} \setminus p$	0%	5%	10%	20%
0 px	**0.859**	—	—	—
2 px	—	0.666	0.599	0.477
5 px	—	0.428	0.386	0.308
10 px	—	0.179	0.162	0.129

perturbed GTs (plus a clean baseline) is generated once with seed 42 and re-evaluated with the same checkpoint; image pixels are untouched. Table XI reports the resulting mAP degradation.

Two patterns emerge. First, *bbox jitter dominates high-IoU precision*: at σ=5 px, AP_{50} drops only modestly (-0.141) whereas AP_{75} collapses (-0.596), since a 5-px center shift on $\sim$100-px boxes breaks IoU$\geq$0.75 but not IoU$\geq$0.50. Second, *class flips degrade* mAP *approximately multiplicatively*: doubling the flip rate roughly halves the remaining mAP at fixed σ. Even at the most adversarial setting, the model retains a $3\times$ margin over the $1/24$ random-class baseline, indicating the learned RF features do not collapse under annotation corruption.

b) Cross-dataset zero-shot transfer: We evaluate the same Dataset 2-trained checkpoint on Dataset 1's test set (220 images, 10,262 instances, 24 shared classes, no fine-tuning). The overall mAP drops to 0.157 (AP_{50}=0.513, AP_{75}=0.039): *coarse localization transfers but precise regression does not*, since AP_{50} remains useful while AP_{75} collapses. At the class level, 14 of 24 classes retain $\mathrm{AP}_{50} > 0.5$ (top: B4 0.931, A3 0.911, A2 0.903), but the C-group chromosomes (C6–C12) fail almost completely ($\mathrm{AP}_{50} < 0.02$). The C-group are acrocentric and morphologically similar, distinguished mainly by subtle banding patterns — the cross-cohort gap is concentrated exactly where intra-cohort difficulty is highest. This is corroborated by k=10-shot fine-tuning on Dataset 1 (§IV-G), which lifts mAP only to 0.055, indicating cohort-matched training data remains essential for the fine-grained head.

V. ANALYSIS AND DISCUSSION

a) RF, Stochastic Coupling, and solver choice.: RF's straight-line ODE paths reduce truncation error in few-step inference, which compounds across the $\sim$46 objects per image; the $+0.082$ mAP gain on Dataset 2 confirms RF's effectiveness in this high-density, small-data, fine-grained regime. Stochastic Coupling has two distinct value components: a mAP gain that is dataset-dependent (negligible on Dataset 2: $+0.0001$, p=0.80; large on Dataset 1: $+0.034$, $p < 10^{-120}$, Table VIII) and a $4.6\times$ epoch-std reduction ($0.006 \to 0.0013$) that holds on both, consistent with the theory that more data attenuates OT collapse. The smoothness benefit aids checkpoint selection — Random coupling's EarlyStopping may land on a "lucky" epoch (0.006 above trend) that fails on test. Because A2 (Heun) and A3 (DPM-Solver++) share identical FM training, weights are identical at each epoch; yet DPM-Solver++ at matched 4-step yields a small but significant mAP improvement ($+0.006$ per-image, Wilcoxon $p < 10^{-3}$; Table VII), so the higher-order solver is slightly *better*, not worse, at 43% fewer NFE — refining FlowDet's conclusion.

b) Theory applicability.: The two-sided bound (Propositions 1–2) is tight in chromosome detection (0.03% gap): well-separated Voronoi cells ($> 20\,\mathrm{px}$ vs $\sigma \sim 1\,\mathrm{px}$) make $P_{\mathrm{err}} < 10^{-45}$. The theory does not transfer to high-dimensional generation ($d \sim 10^5$, $\Delta H/H \approx 0$) nor densely overlapping targets; for COCO ($K \sim 7$, $\Delta H/H \approx 0.55$) it predicts smaller-magnitude benefit.

VI. Conclusion

We introduced *KaryoFlow*, a diffusion-based detector for chromosome karyotyping that brings Rectified Flow into clinical cytogenetics. The RF paradigm — straight-line ODE paths replacing curved DDPM trajectories — is the dominant accuracy source ($+0.076$ mAP over DiffusionDet, matching Cascade R-CNN; 94% of the gain attributed to RF itself). Stochastic Coupling, grounded in our OT Diversity Collapse characterization in the low-dimensional ($\mathbb{R}^4$) detection space, restores coupling diversity and stabilizes training: large mAP gains in low-data regimes ($+0.034$, $p < 10^{-120}$) and a $4.6\times$ epoch-oscillation reduction on larger data. DPM-Solver++ with Top-K pruning delivers four-step inference at 13.3–14.2 FPS with a small but significant precision advantage over Heun, positioning the detector for interactive clinical screening.

The OT Diversity Collapse phenomenon is not chromosome-specific — it arises whenever the prediction space is low-dimensional, target density per image is high, and the training corpus is small. This profile recurs in cell detection (histopathology), lesion detection (mammography, retinal imaging), and colony counting; Table II gives an a-priori diagnostic ($d \ll 100$, $K \gg 10$). Limitations: (i) validation is confined to chromosome data; (ii) Stochastic Coupling's mAP gain is dataset-dependent (the $4.6\times$ stability benefit holds independently); (iii) the theory assumes well-separated targets.

Appendix A
Proofs and Additional Ablations

Correspondence. Supports Section III-C (Propositions 1–3) and Sections III-A, IV-C. Full proofs in the arXiv companion.

Prop. 1 ($\Delta H \leq \log K$). Random coupling: $H_{\mathrm{rand}}(V|X_t) \leq \log K$; OT makes V deterministic (Lemma 1), so $H_{\mathrm{OT}} = 0$.

Prop. 2 (Fano). Fano on the K-way NN classifier: $H_{\mathrm{OT}}(V|X_t) \leq h(P_{\mathrm{err}}) + P_{\mathrm{err}} \log K$, $P_{\mathrm{err}} \leq \binom{K}{2}\Phi(-d_{\min}/(2\sigma_t))$; as $\sigma_t/d_{\min} \to 0$, $\Delta H \to \log K$.

Prop. 3 (envelope theorem). Sinkhorn $\mathcal{V}(\epsilon)$ is concave in ϵ, so $dH/d\epsilon \geq 0$; endpoints give $H \to 0$ and $H \to \log K$.

AdaLN-Zero and shifted schedule. Both null (ΔmAP$=$ 0.000, -0.001); Y chromosome hardest (AP$=$0.779, $\sim$1,803 vs $\sim$7,000 samples per autosome).

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